Henry Bonomolo

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Objective

To apply my technical expertise, problem-solving skills, and leadership in software engineering to advance my studies and career in STEM fields.

Education

Aug 2024 – Present
Anticipated Graduation May 2027, *University of California-Berkeley, Berkeley, California*
- Major: Computer Science, BA (College of Computing, Data Science and Society); GPA: 3.668
- Relevant coursework includes: Math 54, CS61B, CS61A, UGBA 10X & ENVDES 5, CS188, CS70

Aug 2020 – May 2024
Graduated, *Catalina Foothills High School, Tucson, Arizona*
- Graduation May 2024; Weighted GPA: 4.212; Unweighted GPA: 3.935
- Relevant coursework includes: AP Chemistry, AP Physics C (4, 4), AP Government, AP Statistics (4), AP Calculus BC (5), Honors Spanish 5: Film, AP Computer Science A (5), AP Computer Science Principles (5), AP World History

Technical Experience

01/15/2025 - Present
React Native Developer, *Mobile Developers of Berkeley, Berkeley, California*
- Trained through an intensive 10-week training course covering React Native, Swift, and applied mobile development, with a focus on integrating machine learning and database design for advanced app functionality.

08/25/2024 - 05/30/2025
Solar Data API Development Intern, *Energesque, Los Angeles, California*
- Research API options for solar irradiance data and develop a Java program to interpret the data from the API.
- Developed an algorithm to convert JSON data from an API into a CSV file, ensuring compatibility with programs that use solar irradiance data for machine learning model predictions in Los Angeles energy optimization.

05/10/2023 - Present
EKG Research Developer, *Computer and Electrical Engineering, University of Arizona*
- Developed an app to help manage stress using an at-home EKG, incorporating breathing strategies for users.
- Built an API to calculate RMSSD from a heart rate monitor which is currently preparing for human trial testing.
- Worked in a 3 person team to develop an entire android app system with EKG monitor for human patients.

08/5/2020 - 05/10/2024
President & Lead Programmer, *First Tech Challenge, Catalina Foothills High School*
- Led a 20-person team to a 1st place finish at the State Championship with the best autonomous for 3 years.
- Devote 10+ hours a week to writing much of the code for autonomous and driver-controlled robot games in Java.

Primary Projects

01/12/2023 - Present
Spike Inclusive, *React Native, Bluetooth Low Energy (BLE), Async Storage*
- iOS iPad app helping 5+ students with visual and cognitive disabilities participate in classroom robotics.
- Developed Bluetooth command functionality to communicate with Spike Prime robots in a custom-built app.
- Published on the App Store and implemented in Catalina Foothills School District Curriculum.

01/10/2024 - Present
ARES, *Python, ReactJS, Azure Cosmos Database development*
- Developed a website and Discord bot to give First Tech Challenge members real-time, up-to-date statistical information on global teams. The platform is freely available for public download at ares-bot.com.
- ARES backend is hosted on Azure, enabling scalable real-time data processing & querying with API integration.

01/15/2025 - 05/23/2025
FitRealm, *React Native, Artificial Intelligence, Supabase*
- Developed an app to gamify going to the gym with community-driven engagement, where users set goals, earn rewards, and connect with others while maintaining motivation in their fitness journey.
- A social media fitness RPG powered by the OpenAI API, generating personalized workout plans, quests, and challenges to motivate your community and guilds users are able to join.

08/02/2023 - 11/10/2023
Hoax Hunter, *Swift, iOS App Development*
- Developed a machine learning model using a custom dataset sourced from various platforms, including over 100+ curated articles by Create ML, to significantly enhance the model's accuracy and performance.
- Fully functioning social media and fake news detecting app for fact sharing published on the Apple App Store.

Leadership Experience and Involvement

08/10/2020 - Present
Coach & Judge, *First Lego League, Sunrise Drive Elementary*
- Mentored two teams to first-place finishes in programming and building during their weekly FLL meetings, offering hands-on mentorship and technical support to prepare them for both regional and state competitions.

03/20/2023 - 10/15/2024
Lifeguard and Swim Instructor, *City of Tucson, Amphitheater High School, Tucson, Arizona*
- Guided young students in water competency, teaching essential swimming techniques and safety skills.
- Honed responsibility, leadership, and empathy while helping students gain confidence in water skills and safety.

10/15/2022 - 12/20/2022
Volunteer, *First Lego League, Boys and Girls Club*
- Piloted and started the First program at Doolen Middle School. Helped set up supplies and guide instructors.

05/06/2022 - 08/10/2022
Teacher, *Manyatta Primary School, Naru Moru, Kenya*
- Taught various subjects to 30 K-8 students with continuing support by managing fundraisers for Project Rafiki.

Awards & Achievements

- **National Hispanic Scholar** (2023), **AP Scholar Award With Distinction** (2022, 2023, 2024) – Awarded for high scores on various College Board exams.
- **Arizona Gold Seal of Biliteracy: Spanish** (2023) – Earned for demonstrating proficiency in reading, writing and speaking both English and Spanish.
- **Oracle Java Certified Foundations Associate** (2023) – Validated my strong proficiency in Java programming fundamentals through certification.
- **FIRST LEGO League Mentor and Coach Award** (2023) – Recognized for mentoring and coaching teams to success in robotics competitions.

Skills & Abilities

- **Programming Languages**: Proficient in **Java**, **Python, Scheme, SQL, React Native**, and **Swift**, with web development experience in **HTML** and **CSS**.
- **Libraries & Frameworks**: Skilled in data analysis with **Numpy** and **Pandas**; adept in **JSON** processing (javax.json) and familiar in NodeJS.
- **Tools & Platforms**: Proficient in **GitHub**, **Firebase**, **MongoDB**, **AWS RDS, Azure Cosmos, & Supabase** for authentication, and cloud databases.